Egyptian Pharmacy AI Subagent

A Retrieval-Augmented Medication Guidance System

Graduation Project Report

June 2026

Abstract

This report presents the design, methodology, and evaluation of a pharmacy-focused AI subagent built for the Egyptian drug market. The system addresses a concrete gap in digital healthcare: patients seeking medication guidance are frequently exposed to inaccurate information, including hallucinated drug names and prices produced by general-purpose language models unaware of the Egyptian drug registry. The proposed system combines a curated pharmaceutical dataset, a three-channel hybrid retrieval pipeline, a clinical safety screening layer, and a large language model to deliver grounded, hallucination-resistant medication guidance in Egyptian Arabic. Designed as a specialised subagent within a broader voice-agent healthcare workflow, the system strictly confines itself to pharmacy tasks and escalates all out-of-scope requests to the primary orchestrator.

Contents

1 Introduction

1.1 Motivation

Access to accurate medication information is a significant challenge in Egypt. Patients and caregivers routinely seek guidance for self-limiting conditions and over-the-counter (OTC) purchases, yet the advice available to them is inconsistent in accuracy, lacks price transparency, and is rarely tailored to individual contraindications or allergy profiles.

The rise of general-purpose AI assistants has not resolved this problem — it has compounded it. Large language models trained on broad internet corpora can produce confident, fluent responses to medication queries, but they frequently hallucinate drug names, invent prices, and reference products that are not registered in the Egyptian market. A patient acting on such a response risks purchasing a non-existent product, taking an unsafe dose, or missing a contraindication relevant to their health profile.

This project was motivated by the need for a medication guidance system that is both conversationally capable and factually grounded — one that draws every drug name, price, and clinical warning directly from a verified Egyptian pharmaceutical dataset rather than from a language model's parametric memory.

1.2 Problem Statement

The core problem addressed by this project is the lack of a reliable medication assistant tailored to the Egyptian pharmaceutical market. General-purpose AI systems may provide medication-related information, but they are not grounded in local drug data and may generate inaccurate or unsupported responses.

This project aims to provide a pharmacy-focused assistant that delivers accurate drug information, prices, active ingredients, and substitute recommendations based on a curated Egyptian medication dataset. The system follows a retrieval-first approach, ensuring that medication information originates from verified dataset records rather than model-generated content.

In addition, the assistant is intentionally limited to pharmacy-related tasks. It does not perform medical diagnosis, treatment planning, appointment booking, or emergency triage. Requests that fall outside the pharmacy domain are delegated to the main healthcare orchestrator, ensuring clear separation of responsibilities and safer user interactions.

1.3 Design Philosophy

The system was designed as a specialized pharmacy assistant focused on medication-related queries. Instead of relying on a large language model as the primary source of information, the architecture follows a retrieval-first approach where medication data is retrieved from a curated Egyptian drug dataset before any response is generated.

This design was chosen to improve reliability when handling medication names, active ingredients, prices, and substitute recommendations. By separating information retrieval from response generation, the system reduces the risk of hallucinated drug information and ensures that pharmaceutical facts originate from structured dataset records.

The language model is therefore used as a supporting component for conversational guidance and response formulation, while medication information remains grounded in the retrieval layer.

1.4 Why Retrieval-Augmented Generation

A pure language model approach was evaluated and rejected at the design stage. Table 1 summarises the key limitations of standalone LLMs and how Retrieval-Augmented Generation (RAG) addresses each.

Table 1: Rationale for RAG over a standalone LLM

LLM Limitation	How RAG Addresses It
Drug-name hallucination	Retrieval is performed against a verified dataset; the LLM is prohibited from writing drug names in its output.
No Egyptian market knowledge	The dataset contains only drugs registered in Egypt, with Arabic trade names and EGP prices.
No price accuracy	Prices are retrieved from the dataset on every query, not generated by the model.
No auditability	Every drug fact is tied to a specific dataset record.
Unsafe safety enforcement	Contraindication checks require structured drug facts that only retrieval can supply; model judgment is insufficient.

RAG separates *factual grounding* (retrieval from the verified dataset) from *language generation* (conversational explanation by the model). The model's role is strictly to produce natural-language usage guidance around facts that have already been retrieved and safety-checked.

2 Dataset Preparation

2.1 Dataset Collection

The pharmaceutical dataset was collected through web scraping from EgyptDwa, a publicly accessible Egyptian drug information portal that provides medication information available in the Egyptian market.

The collected records include medication trade names (Arabic and English), active ingredients, dosage forms, manufacturers, package information, and listed retail prices. Following the data collection phase, the dataset underwent several preprocessing steps, including data cleaning, normalisation, duplicate handling, and price correction, to improve data quality and consistency.

The resulting dataset serves as the primary knowledge source for the retrieval system. All medication information presented by the pharmacy assistant is retrieved from this curated dataset rather than generated directly by the language model, ensuring greater factual consistency and reducing the risk of hallucinated drug information.

2.2 Data Cleaning and Normalisation

Raw scraped data required a structured cleaning pipeline before use in retrieval. The key stages were:

- **Missing value handling** — Records lacking a trade name, active ingredient, or dosage form were removed or enriched from secondary sources. Fields critical to retrieval and safety screening were treated as required.
- **Duplicate removal** — Exact and near-duplicate records were identified and eliminated. Near-duplicate detection addressed entries differing only in whitespace, punctuation, or Arabic orthographic variation, including Hamza normalisation, Taa Marbouta, and Alef Maqsoura equivalences.
- **Active-ingredient normalisation** — Ingredient strings were resolved to canonical International Non-proprietary Names (INNs). Common synonyms — such as *acetaminophen* to *paracetamol* — were unified, and compound ingredients were parsed into component tokens to support ingredient-level safety screening.
- **Dosage normalisation** — Numeric strength and unit were standardised and extracted into dedicated fields, enabling constraint-based numeric matching during retrieval.
- **Bilingual name normalisation** — Arabic names were normalised for common script variants; English names were lowercased and stripped of trailing form noise to improve matching consistency.

- **Ingredient alias mapping** — A curated synonym dictionary maps colloquial, regional, and alternative spellings to canonical INNs, so that queries using brand names or informal names resolve correctly to the underlying active ingredient.

2.3 Price Validation and Correction

Some records scraped from `egyptdwa.com` contained anomalous or unrealistic price values, including zero-value entries, implausibly low figures, and extreme outliers attributable to data-entry errors. A dedicated price validation and correction stage was introduced to address these issues before the dataset was used for retrieval.

The correction methodology groups similar medications by shared attributes: normalised active ingredient, dosage form, strength, and brand prefix. Within each peer group, the median price serves as the reference value — a statistically robust measure resistant to extreme outliers. Entries deviating substantially from their peer-group median are assigned a corrected price derived from that median. The original scraped price is retained alongside the corrected value in every record, preserving full auditability.

Table 2 illustrates the scale of corrections applied to a representative sample of affected records.

Table 2: Representative price corrections applied by the validation framework

Drug	Active Ingredient	Before (EGP)	After (EGP)	Change
Immunomide 5 mg 21 Caps.	lenalidomide	6,998	56,702	+710%
Immunomide 25 mg 21 Caps.	lenalidomide	8,404	56,702	+575%
Otathromb 75 mg 10 Tabs.	clopidogrel	9	48	+433%
Powerecta 5 mg 1 Tab.	vardenafil	3	30.50	+917%

3 System Architecture

3.1 Overview

The system is structured as a layered pipeline in which each stage has a single, well-defined responsibility. Queries enter through a REST API and pass through query understanding, hybrid retrieval, safety screening, and response grounding before a structured response is returned to the calling orchestrator. Figure 1 illustrates the end-to-end flow.

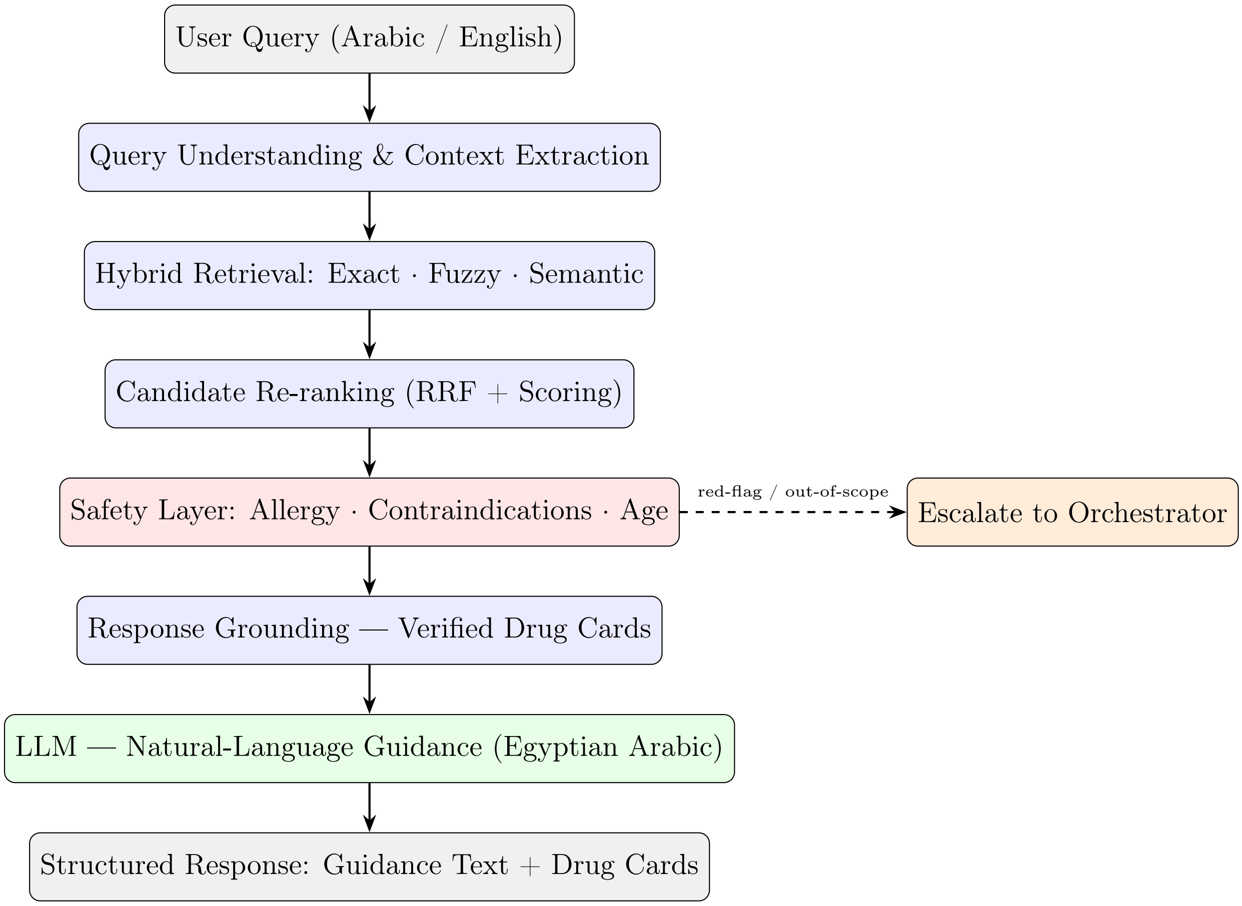

Figure 1: End-to-end system pipeline

3.2 Main Components

Query Understanding

Each incoming message is parsed to extract a structured patient profile covering age, sex, pregnancy and breastfeeding status, declared allergens, chronic conditions, current medications, and symptom description. The query is simultaneously classified by intent: a product information request, a substitute request, or an OTC symptom-treatment query. Patient context is accumulated across conversation turns so that information declared early in the dialogue continues to govern drug selection in later turns.

Hybrid Retrieval Engine

A three-channel retrieval engine searches the pharmaceutical dataset using complementary matching strategies. Candidates from all channels are merged and re-ranked before safety screening. The three channels — exact matching, fuzzy matching, and semantic vector search — are described in detail in Section 4.

Safety Layer

Every candidate drug is screened against the patient's declared allergens, chronic conditions, pregnancy or breastfeeding status, and age. Contraindicated candidates are suppressed before the LLM is invoked. Queries triggering clinical red flags — such as chest pain, severe dyspnoea, or loss of consciousness — are immediately escalated to the orchestrator without drug retrieval.

Response Grounding
Candidates that pass safety screening are assembled into structured drug cards drawn directly from dataset fields: Arabic and English trade names, active ingredient, dosage form, and price. These cards are passed to the LLM as verified context and are also returned as structured objects in the API response, independent of the model's generated text.

LLM Guidance Layer
The language model receives the patient's message, conversation history, patient context, and the pre-assembled drug cards. Its role is narrowly defined: produce natural-language usage guidance and clinical warnings in colloquial Egyptian Arabic. It is explicitly prohibited from writing drug names, prices, or active ingredients — all of which appear only in the structured cards.

Workflow Envelope
Every response includes a structured workflow envelope carrying a task status, a flag indicating whether control should return to the orchestrator, and an escalation reason where applicable. This envelope enables the calling orchestrator to make routing decisions without parsing the natural-language text.

3.3 Data Flow

A typical OTC query follows this path through the system. The patient's message and conversation history are received by the API. The query understanding layer extracts the patient's context and classifies the intent. The retrieval engine searches the dataset across three channels and merges the results. The safety layer screens the merged candidates and suppresses contraindicated drugs. Surviving candidates are assembled into drug cards, which are passed to the LLM together with the patient context. The LLM produces guidance text in Egyptian Arabic. The final response combines the guidance text, the drug cards, and the workflow envelope, and is returned to the orchestrator.

For product information queries — where the user asks about a specific named drug — the retrieval step targets trade name and active ingredient fields directly, and no clinical safety gate is applied since the user is seeking information rather than a recommendation.

4 Retrieval-Augmented Generation Pipeline

4.1 Query Understanding

Before retrieval begins, the system classifies the query and builds a patient context record from the full conversation history. Classification distinguishes three query types: product information requests (questions about a named drug's price, ingredients, or details), substitute requests (requests for a therapeutically equivalent alternative), and symptom-based OTC queries (requests for a drug recommendation based on described symptoms). Each type activates a different retrieval strategy and different mandatory safety checks.

Patient context extraction accumulates structured information across turns: age and sex determine age-gating and pregnancy-check applicability; declared allergens govern allergy screening; chronic conditions activate contraindication rules; and red-flag symptom patterns trigger immediate escalation. Because context is maintained across the conversation, a patient who mentions a chronic condition early in the dialogue will have that condition respected for every subsequent drug suggestion.

4.2 Hybrid Retrieval

The retrieval engine operates over three complementary channels.

Exact name matching performs deterministic string matching against the full set of Arabic and English trade names. A match is declared when the normalised query equals, begins with, or is contained within a normalised drug name. Exact matches receive a scoring bonus that prevents them from being displaced by weaker approximate hits.

Fuzzy matching applies an ensemble of five similarity algorithms (ratio, partial ratio, token sort ratio, token set ratio, and weighted ratio). The highest score across all five is retained for each candidate, making the approach more robust than any single algorithm against prefix matches, word reordering, and partial abbreviations. Separate fuzzy passes target the active ingredient field, the combined name-plus-ingredient field, and the Arabic and English name fields independently, each with tuned minimum thresholds.

Semantic vector search performs approximate nearest-neighbour search over pre-computed multilingual sentence embeddings using a FAISS flat inner-product index. This channel enables natural-language queries — for example, *"something for a blocked nose in children"* — to match relevant entries without any token overlap. A multilingual MiniLM model handles bilingual Arabic/English queries. The embedding index is built offline and loaded from disk at startup, so only a single short embedding call per query is required at runtime, keeping memory usage low.

4.3 Candidate Re-ranking

Candidates from all three channels are merged into a unified pool. A Reciprocal Rank Fusion (RRF) mechanism boosts candidates that appear in more than one channel, proportional to the number of channels that returned them. The composite scoring function additionally incorporates ingredient-overlap score, dosage form preference, and numeric strength compatibility. A relevance gate is applied after re-ranking: substitute and ingredient-mode queries require strict pharmacological confirmation before a candidate is passed to the safety layer.

4.4 Safety Validation

The safety layer applies four categories of constraint before any drug is surfaced to the user.

Allergy screening tokenises declared allergens and compares them at the component level against the active ingredient tokens of every candidate. A candidate is suppressed if any allergen token matches any component of its active ingredient string — an approach that correctly handles compound ingredients.

Contraindication checking matches the patient's chronic conditions against a structured knowledge base of ingredient-level contraindication rules. NSAIDs, for example, are contraindicated in hypertension, kidney disease, peptic ulcer, and diabetes; decongestants containing pseudoephedrine are contraindicated in hypertension and heart disease. Candidates violating any absolute rule are removed from the pool.

Pregnancy and breastfeeding restrictions block NSAIDs, certain decongestants, and flagged antihistamines for patients who are pregnant or breastfeeding. When pregnancy status is unknown for a female patient of childbearing age and the candidate pool contains any flagged ingredient, the system requests confirmation before proceeding.

Age gating suppresses molecules that carry minimum-age restrictions for patients who fall below the safe threshold.

4.5 LLM Guidance Generation and Response Grounding

Candidates that pass all safety checks are assembled into structured drug cards containing Arabic and English trade names, active ingredient, dosage form, and price — all drawn directly from dataset fields. These cards are passed to the language model as verified context. The model's system prompt explicitly instructs it that drug details are displayed in separate cards and must not be reproduced or paraphrased in the generated text; its sole task is to produce usage guidance and warnings in colloquial Egyptian Arabic.

As a defence-in-depth measure, a post-processing step scans the model's output and removes any drug names, prices, or ingredient strings that appear verbatim, replacing them with neutral phrasing. This ensures that even if the model violates the system prompt constraint, no hallucinated drug facts propagate to the final response. The drug cards are returned to the orchestrator as structured JSON objects independently of the guidance text, allowing the client to render them even if the text is modified or replaced.

5 Evaluation and Results

5.1 Evaluation Methodology

The system was evaluated across three dimensions using a structured test suite of curated trade-name, active-ingredient, and substitute queries, including intentionally misspelled variants to validate fuzzy robustness.

Active Ingredient Preservation measures whether every substitute returned for a source drug shares the identical active ingredient. This is the primary pharmacological correctness criterion: a substitute that changes the active ingredient is not a safe alternative.

Dosage Form Preservation measures whether every returned substitute shares the same dosage form as the source drug (e.g. tablet, syrup, capsule). A form change alters the route of administration and may be clinically inappropriate.

Constraint Preservation measures the accuracy of the numeric constraint filter — specifically, whether the system correctly passes candidates matching the queried dosage strength and correctly suppresses those that do not.

5.2 Results

Table 3: System evaluation results

Metric	Score	Interpretation
Active Ingredient Preservation	100%	All returned substitutes share the exact active ingredient of the source drug.
Dosage Form Preservation	100%	No substitute changes the dosage form or route of administration.
Constraint Preservation	86%	Numeric dosage-strength filtering is accurate in the majority of cases; residual errors arise from non-standard strength notations.

5.3 Discussion

The 100% scores on Active Ingredient and Dosage Form Preservation reflect the grounding architecture's core design: substitutes are selected exclusively by the retrieval engine using pharmacological constraints, and the language model is never permitted to propose alternatives of its own. These results confirm that the system does not hallucinate therapeutically inequivalent suggestions.

The Constraint Preservation score of 86% identifies numeric dosage-strength matching as the principal remaining challenge. Strength values in the dataset are expressed in a variety of notations — mg, mg/ml, mg/5 ml, % — and standard string-similarity approaches conflate different concentrations. The current implementation uses explicit numeric parsing to handle the most common cases, but non-standard notations remain a source of error. Improving this layer represents the clearest path to near-perfect constraint preservation.

6 Challenges

6.1 Dataset Quality and Price Inconsistency

The primary data source contained a non-trivial proportion of records with anomalous or unrealistic prices, arising from data-entry errors and crowd-submitted corrections on the source portal. A dedicated price validation framework — described in Section 2.3 — was developed to detect and correct these outliers without discarding the original data. Ensuring dataset quality was an iterative process that required repeated manual inspection alongside automated statistical correction.

6.2 Arabic and English Naming Variation

Egyptian drug names appear in a wide range of forms: full Arabic trade names, Arabicised transliterations, abbreviated English brand names, and informal colloquial references. A query for *"panadol"* must resolve to paracetamol; a query in Arabic script must match the same entry as one in English. The hybrid retrieval approach — combining exact matching, fuzzy matching, and multilingual semantic search — was adopted specifically to handle this variation. A curated ingredient alias map provides an additional normalisation layer for common brand-to-ingredient mappings.

6.3 Preventing Hallucinated Medication Information

General-purpose language models produce fluent, confident text, which makes their hallucinations particularly dangerous in a medical context. The system addresses this through three complementary mechanisms: (1) the LLM's system prompt explicitly prohibits it from writing drug names, prices, or active ingredients; (2) drug facts are delivered through structured dataset-assembled cards, not generated text; and (3) a post-processing sanitisation step removes any drug-specific strings that appear in the model's output. The defence-in-depth approach ensures that no hallucinated claim reaches the user even if the model partially violates its instructions.

6.4 Memory Constraints at Deployment

The multilingual sentence-transformer model required for semantic vector search consumes significant memory. On the constrained hosting environment used for deployment, loading the model and encoding the full drug dataset at startup caused out-of-memory failures. This was resolved by decoupling index construction from serving: the embedding index is built offline once using a dedicated build script, then committed to the repository and loaded from disk at runtime. At serving time, only a single short query embedding call per request is required, reducing peak memory consumption to within the available limit.

7 Future Work

Several concrete directions are identified for extending the system.

Improved numeric constraint matching. The 14% gap in Constraint Preservation represents the most immediate improvement target. Developing a more comprehensive numeric normalisation layer that handles all common strength notation variants would bring this metric close to the 100% achieved on pharmacological correctness metrics.

Drug–drug interaction checking. The current system screens a candidate drug against the patient's chronic conditions and allergies, but does not cross-reference it against the patient's current medications. Adding a structured interaction checker that compares candidate active ingredients against declared current medications would significantly strengthen the clinical safety layer.

Expanded dataset coverage. The current dataset reflects a single scraped source. Expanding coverage to additional Egyptian pharmacy portals and drug-authority registries would improve both completeness and the accuracy of substitute recommendations, particularly for the 8% of source drugs for which no substitute is currently found.

Quantised vector index. Replacing the current flat FAISS index with a quantised index (IVF or product quantisation) would reduce memory consumption substantially while preserving recall, enabling semantic search to be activated on more constrained hosting environments without out-of-memory risk.

Learned re-ranking. The current re-ranking formula is hand-tuned and heuristic. Training a lightweight re-ranker model on pharmacy query–result pairs would produce a more accurate and generalisable candidate ordering, particularly for ambiguous queries that span multiple candidate drug families.

8 Conclusion

This project delivers a production-deployed pharmacy AI subagent grounded in the Egyptian drug market. The system demonstrates that the combination of a verified pharmaceutical dataset, hybrid retrieval, clinical safety screening, and a strictly constrained language model can produce medication guidance that is both conversationally natural and factually safe — without relying on the language model's parametric memory for any clinical fact.

The evaluation results confirm the core architectural claim: by separating factual retrieval from language generation, and by enforcing pharmacological constraints at the retrieval stage rather than the generation stage, the system achieves 100% preservation of active ingredient and dosage form correctness across all tested substitute queries. The principal remaining challenge — numeric dosage-strength constraint matching at 86% — is well-understood and represents a clear path for future improvement.

The subagent design pattern proved to be the right architectural choice for this domain. A specialised agent with a narrow, well-defined scope is significantly safer, more auditable, and more maintainable than a monolithic general-purpose medical chatbot. The structured workflow envelope enables seamless integration into larger healthcare orchestration systems, ensuring that out-of-scope requests — diagnosis, emergency triage, appointment booking — are handled by the appropriate component rather than the pharmacy agent.

— End of Report —